

The sharp universal spectral constant for scaled q -numerical ranges

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Abstract

Let A be an $n \times n$ complex matrix, $n \geq 2$, and let $q \in \mathbb{C}$ satisfy $0 < |q| \leq 1$. The q -numerical range and its scaled form are

$$W_q(A) = \{y^*Ax : \|x\|_2 = \|y\|_2 = 1, y^*x = q\}, \quad \Omega_q(A) = q^{-1}W_q(A).$$

We prove that the sharp universal spectral constant for the family of scaled q -numerical ranges is

$$C_q = \max \left\{ 1, \frac{2|q|}{1 + \sqrt{1 - |q|^2}} \right\}.$$

Thus, for every rational function f pole-free on $\Omega_q(A)$, $\|f(A)\|$ is at most C_q times the maximum modulus of f on $\Omega_q(A)$, and C_q is the least constant valid uniformly in n and A . The proof is a transfer from the sharp numerical-range theorem. Its geometric input shows that the numerical range of every adapted rank-one stretch of A lies in the scaled q -range. Applying a numerical-range spectral-set estimate to these stretches produces a simultaneous family of Schur estimates. A disk automorphism adapted to a maximizing singular-vector pair then extracts a norm bound for the original matrix. More generally, a universal numerical-range spectral constant K transfers to the constant $\max\{1, K/\kappa(q)\}$ on the scaled q -range, where $\kappa(q) = (1 + \sqrt{1 - |q|^2})/|q|$. The two lower-bound branches defining C_q are attained, respectively, by the constant rational function and by the identity polynomial on a 2×2 square-zero matrix.

Keywords: q -numerical range; spectral set; Crouzeix conjecture; numerical range; rank-one similarity; Schur function

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1. Introduction

For $A \in \mathbb{M}_n(\mathbb{C})$, its numerical range is

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}.$$

It is compact and convex, and it contains the spectrum of A . For vectors, $\|\cdot\|_2$ denotes the Euclidean norm. For a matrix T , the unsubscripted norm is the induced Euclidean operator norm,

$$\|T\| := \max_{\|x\|_2=1} \|Tx\|_2.$$

For $C \geq 1$, a compact set $E \subset \mathbb{C}$ containing $\sigma(A)$ is a C -spectral set for A if

$$\|g(A)\| \leq C \max_{z \in E} |g(z)| \tag{1}$$

for every rational function g pole-free on E . Crouzeix conjectured that $W(A)$ is always a 2-spectral set [1], and the sharp constant 2 was established in [2].

Throughout, $q \in \mathbb{C}$. For $n \geq 2$ and $0 < |q| \leq 1$, the q -numerical range is

$$W_q(A) = \{y^*Ax : \|x\|_2 = \|y\|_2 = 1, y^*x = q\}, \quad \Omega_q(A) = \frac{1}{q}W_q(A). \quad (2)$$

The set $W_q(A)$ is compact, and Tsing proved its convexity [3]. The scaled set $\Omega_q(A)$ contains $W(A)$ and hence $\sigma(A)$. O’Loughlin and Rani formulated the polynomial inequality below as Conjecture 4.2 [4]. The theorem proves that conjecture together with its rational spectral-set extension.

Theorem 1.1 (Main theorem). *Let $n \geq 2$, let $A \in \mathbb{M}_n(\mathbb{C})$, let $0 < |q| \leq 1$, and let f be a rational function pole-free on $\Omega_q(A)$. Then*

$$\|f(A)\| \leq C_q \max_{z \in \Omega_q(A)} |f(z)|, \quad C_q = \max \left\{ 1, \frac{2|q|}{1 + \sqrt{1 - |q|^2}} \right\}. \quad (3)$$

Equivalently, $\Omega_q(A)$ is a C_q -spectral set for A . In particular, the same inequality holds for every polynomial f .

The proof has three steps. First, Section 2 proves nesting of the scaled ranges and shows that every adapted rank-one stretch has numerical range inside $\Omega_q(A)$. Second, Section 3 proves that uniform Schur estimates over precisely these stretches force an improved norm estimate for the original matrix. Third, Section 4 packages the two facts into a transfer principle for any universal numerical-range spectral constant K . Taking $K = 2$ gives (3); a 2×2 nilpotent matrix proves sharpness.

2. Rank-one stretch geometry

We use the inner product

$$\langle x, y \rangle = y^*x,$$

which is linear in its first variable. Thus $W_q(A) = \{\langle Ax, y \rangle : \|x\|_2 = \|y\|_2 = 1, \langle x, y \rangle = q\}$. Changing the phase of y in (2) gives

$$\Omega_q(A) = \Omega_{|q|}(A). \quad (4)$$

We may therefore put

$$r = |q|, \quad \tau = \sqrt{1 - r^2}, \quad \kappa = \frac{1 + \tau}{r}. \quad (5)$$

The equivalent identities

$$r = \frac{2\kappa}{\kappa^2 + 1}, \quad \frac{\tau}{r} = \frac{\kappa^2 - 1}{2\kappa}, \quad \frac{2}{\kappa} = \frac{2r}{1 + \sqrt{1 - r^2}} \quad (6)$$

will be used repeatedly.

For $0 < r < 1$, division by r in Tsing’s Lemma 5 gives the disk formula below [3]. The endpoint $r = 1$ is included in the proof. The formula will also be used in the sharpness argument. For a unit vector x , put

$$\rho_A(x) = \left(\|Ax\|_2^2 - |\langle Ax, x \rangle|^2 \right)^{1/2}.$$

The radicand is nonnegative by Cauchy–Schwarz.

Lemma 2.1 (Disk formula and nesting). *Let $A \in \mathbb{M}_n(\mathbb{C})$, $n \geq 2$, and $0 < r \leq 1$. Then*

$$\Omega_r(A) = \bigcup_{\|x\|_2=1} \left\{ z \in \mathbb{C} : |z - \langle Ax, x \rangle| \leq \frac{\sqrt{1 - r^2}}{r} \rho_A(x) \right\}. \quad (7)$$

Consequently, if $0 < r \leq s \leq 1$, then

$$\Omega_s(A) \subseteq \Omega_r(A).$$

In particular, $W(A) = \Omega_1(A) \subseteq \Omega_r(A)$.

Proof. Let x, y be unit vectors with $\langle x, y \rangle = r$, and put $e_x = Ax - \langle Ax, x \rangle x$. Then $e_x \perp x$ and $\|e_x\|_2 = \rho_A(x)$. Since $y - rx \perp x$ and $\|y - rx\|_2 = \sqrt{1 - r^2}$, Cauchy–Schwarz gives

$$\left| \frac{1}{r} \langle Ax, y \rangle - \langle Ax, x \rangle \right| = \frac{1}{r} |\langle e_x, y - rx \rangle| \leq \frac{\sqrt{1 - r^2}}{r} \rho_A(x).$$

This proves the inclusion from left to right in (7).

Conversely, suppose that ζ belongs to the disk indexed by a unit vector x_0 . On the unit sphere define

$$G(x) = |\zeta - \langle Ax, x \rangle| - \frac{\sqrt{1 - r^2}}{r} \rho_A(x).$$

The function G is continuous and $G(x_0) \leq 0$. A complex matrix has a unit eigenvector e , and $\rho_A(e) = 0$, so $G(e) \geq 0$. The unit sphere in $\mathbb{C}^n$ is path connected for $n \geq 2$; hence there is a unit vector u with $G(u) = 0$.

Write $\alpha = \langle Au, u \rangle$ and $d = \rho_A(u)$. Choose a unit vector $w \perp u$ such that $\langle Au, w \rangle = d$: when $d > 0$, take $w = (Au - \alpha u)/d$, and when $d = 0$, take any unit vector orthogonal to u . If $d > 0$, set

$$\beta = \frac{r \overline{\zeta} - \alpha}{d};$$

if $d = 0$, set $\beta = \sqrt{1 - r^2}$. The equality $G(u) = 0$ gives $|\beta| = \sqrt{1 - r^2}$. Thus $y = ru + \beta w$ is a unit vector with $\langle u, y \rangle = r$, and if $d > 0$, the definition of β gives $\bar{\beta}d/r = \zeta - \alpha$. If $d = 0$, then $G(u) = 0$ gives $\zeta = \alpha$. In either case,

$$\frac{1}{r} \langle Au, y \rangle = \alpha + \frac{\bar{\beta}d}{r} = \zeta.$$

This proves the reverse inclusion.

Finally, the coefficient $\sqrt{1 - r^2}/r$ decreases with r on $(0, 1]$. Comparing the disks in (7) proves nesting, and the case $r = 1$ gives $\Omega_1(A) = W(A)$. $\square$

For a unit vector v , let P_v be the orthogonal projection onto $\mathbb{C}v$ and define the rank-one stretch

$$S_v = I + (\kappa - 1)P_v. \quad (8)$$

Lemma 2.2 (Rank-one stretch geometry). *Let $A \in \mathbb{M}_n(\mathbb{C})$, $n \geq 2$, and $0 < r \leq 1$, with κ as in (5). For every unit vector v , $S_v \succ 0$,*

$$S_v^{-1} = I + (\kappa^{-1} - 1)P_v,$$

and

$$W(S_v A S_v^{-1}) \subseteq \Omega_r(A). \quad (9)$$

Proof. The formulas for S_v and S_v^{-1} follow from $P_v^2 = P_v$; positivity follows from $\kappa \geq 1$. Fix a unit vector u and put

$$\ell = \|S_v^{-1}u\|_2, \quad m = \|S_v u\|_2, \quad \theta = \|P_v u\|_2^2.$$

Orthogonality of $P_v u$ and $(I - P_v)u$ gives

$$m^2 = \kappa^2 \theta + 1 - \theta, \quad \ell^2 = \kappa^{-2} \theta + 1 - \theta.$$

Since $0 \leq \theta \leq 1$,

$$\ell^2 m^2 = 1 + (\kappa - \kappa^{-1})^2 \theta(1 - \theta) \leq \frac{1}{4}(\kappa + \kappa^{-1})^2.$$

Moreover, $1 = |\langle S_v^{-1}u, S_v u \rangle| \leq \ell m$. Because S_v is invertible, $\ell, m > 0$. Therefore, with $s = (\ell m)^{-1}$,

$$r \leq s \leq 1, \quad \frac{1}{2}(\kappa + \kappa^{-1}) = \frac{1}{r}.$$

The vectors

$$x = \frac{S_v^{-1}u}{\ell}, \quad y = \frac{S_v u}{m}$$

are unit vectors with $\langle x, y \rangle = s$, and

$$\langle S_v A S_v^{-1} u, u \rangle = \frac{1}{s} \langle A x, y \rangle \in \Omega_s(A) \subseteq \Omega_r(A)$$

by Lemma 2.1. This proves (9). $\square$

3. Rank-one stretch extraction

Write $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ for the open unit disk. A convenient family of disk automorphisms is

$$\varphi_{\omega,a}(z) = \frac{\omega z - a}{1 - a\omega z}, \quad |\omega| = 1, \quad 0 \leq a < 1.$$

The next lemma is independent of numerical ranges. It is the step that converts simultaneous estimates over the rank-one stretches (8) into a sharper norm bound for the original matrix.

Lemma 3.1 (Rank-one stretch extraction). *Let $n \geq 2$, let $\kappa \geq 1$ and $K \in \mathbb{R}$, and let $X \in \mathbb{M}_n(\mathbb{C})$ satisfy $\sigma(X) \subseteq \overline{\mathbb{D}}$. Suppose that*

$$\|S_v \varphi_{\omega,a}(X) S_v^{-1}\| \leq K \tag{10}$$

for every unit vector v , every $|\omega| = 1$, and every $0 \leq a < 1$. Then

$$\|X\| \leq \max\left\{1, \frac{K}{\kappa}\right\}. \tag{11}$$

Proof. Put $t = \|X\|$. The result is immediate if $t \leq 1$, so assume $t > 1$. Choose unit vectors x, y such that

$$Xx = ty.$$

Let $c_0 = \langle x, y \rangle$, put $\omega = c_0/|c_0|$ when $c_0 \neq 0$, and put $\omega = 1$ otherwise. Set

$$\tilde{y} = \omega y.$$

Then $|\omega| = 1$, $\omega Xx = t\tilde{y}$, and $c := \langle x, \tilde{y} \rangle = |c_0| \in [0, 1]$. In fact $c < 1$, because $c = 1$ would give $\tilde{y} = x$ and $Xx = t\bar{\omega}x$, producing an eigenvalue of X of modulus $t > 1$, whereas $\sigma(X) \subseteq \overline{\mathbb{D}}$.

If $c = 0$, put $a = 0$. If $c > 0$, consider

$$F(a) = tc(1 + a^2) - a(1 + t^2).$$

Since

$$F(0) = tc > 0, \quad F(1) = 2tc - (1 + t^2) \leq -(t - 1)^2 < 0,$$

there is an $a \in (0, 1)$ such that

$$tc(1 + a^2) = a(1 + t^2). \tag{12}$$

Define

$$u = (I - a\omega X)x = x - at\tilde{y}, \quad w = (\omega X - aI)x = t\tilde{y} - ax.$$

The matrix $I - a\omega X$ is invertible because $\sigma(X) \subseteq \overline{\mathbb{D}}$ and $a < 1$, so $u \neq 0$ and

$$\varphi_{\omega,a}(X)u = w.$$

Also $w \neq 0$, since $w = 0$ would imply $t = \|Xx\|_2 = a < 1$. Equation (12) gives

$$\langle u, w \rangle = tc(1 + a^2) - a(1 + t^2) = 0.$$

It also yields

$$\begin{aligned} \|w\|_2^2 - t^2\|u\|_2^2 &= a(t^2 - 1)(2tc - a(1 + t^2)) \\ &= atc(t^2 - 1)(1 - a^2) \geq 0. \end{aligned} \tag{13}$$

Consequently,

$$\frac{\|w\|_2}{\|u\|_2} \geq t.$$

Put $v = w/\|w\|_2$. Since $u \perp w$, the stretch S_v satisfies $S_v^{-1}u = u$ and $S_v w = \kappa w$. Testing $S_v \varphi_{\omega,a}(X)S_v^{-1}$ on u gives

$$K \geq \|S_v \varphi_{\omega,a}(X)S_v^{-1}\| \geq \kappa \frac{\|w\|_2}{\|u\|_2} \geq \kappa t.$$

Thus every case with $t > 1$ has $t \leq K/\kappa$, proving (11). $\square$

4. Transfer to scaled q -numerical ranges

We now combine the geometry and extraction into a general transfer principle.

Theorem 4.1 (Spectral-constant transfer). *Let $K \geq 1$ have the following property: for every square matrix B and every rational function h pole-free on $W(B)$,*

$$\|h(B)\| \leq K \max_{z \in W(B)} |h(z)|. \tag{14}$$

Let $n \geq 2$, $A \in \mathbb{M}_n(\mathbb{C})$, $0 < |q| \leq 1$, and let f be a rational function pole-free on $\Omega_q(A)$. With

$$r = |q|, \quad \kappa = \frac{1 + \sqrt{1 - r^2}}{r},$$

one has

$$\|f(A)\| \leq \max\left\{1, \frac{K}{\kappa}\right\} \max_{z \in \Omega_q(A)} |f(z)|. \tag{15}$$

Proof. By (4), it is enough to use $\Omega_r(A)$. Put

$$M = \max_{z \in \Omega_r(A)} |f(z)|, \quad M_\varepsilon = M + \varepsilon, \quad X_\varepsilon = \frac{f(A)}{M_\varepsilon},$$

where $\varepsilon > 0$. The maximum exists because $\Omega_r(A)$ is compact and f is continuous there. Since

$$\sigma(A) \subseteq W(A) \subseteq \Omega_r(A),$$

the rational spectral mapping theorem gives $\sigma(X_\varepsilon) \subset \mathbb{D}$.

Fix a unit vector v and put $B = S_v A S_v^{-1}$. Lemma 2.2 gives

$$W(B) \subseteq \Omega_r(A).$$

For $|\omega| = 1$ and $0 \leq a < 1$, define the rational function

$$h(z) = \varphi_{\omega,a} \left(\frac{f(z)}{M_\varepsilon} \right).$$

The function f is pole-free on $W(B)$, and the strict inequality $M/M_\varepsilon < 1$ keeps its normalized values in $\mathbb{D}$. For every $z \in W(B)$,

$$\left| 1 - \frac{a\omega f(z)}{M_\varepsilon} \right| \geq 1 - \frac{aM}{M_\varepsilon} > 0.$$

Thus h is pole-free on $W(B)$ and $|h| \leq 1$ there. Applying (14) and using similarity covariance and composition covariance of the rational functional calculus, we obtain

$$\|S_v \varphi_{\omega,a}(X_\varepsilon) S_v^{-1}\| = \left\| \varphi_{\omega,a} \left(\frac{f(B)}{M_\varepsilon} \right) \right\| \leq K.$$

Lemma 3.1 therefore gives

$$\frac{\|f(A)\|}{M_\varepsilon} \leq \max \left\{ 1, \frac{K}{\kappa} \right\}.$$

Letting $\varepsilon \downarrow 0$ proves (15). $\square$

The sharp numerical-range theorem gives (14) with $K = 2$ for every rational h pole-free on $W(B)$ [2].

Proof of Theorem 1.1. Apply Theorem 4.1 with $K = 2$ and use (6). This proves (3). $\square$

We next establish optimality of both branches of C_q .

Proposition 4.2 (Sharpness). *For every fixed q with $0 < |q| \leq 1$, put $r = |q|$. The constant*

$$C_q = \max \left\{ 1, \frac{2}{\kappa} \right\}, \quad \kappa = \frac{1 + \sqrt{1 - r^2}}{r},$$

is the least constant C such that

$$\|f(A)\| \leq C \max_{z \in \Omega_q(A)} |f(z)|$$

for every $A \in \mathbb{M}_2(\mathbb{C})$ and every rational function f pole-free on $\Omega_q(A)$.

Proof. Theorem 1.1 shows that C_q is admissible. By (4), $\Omega_q(A) = \Omega_r(A)$. The constant function $f \equiv 1$ gives the lower bound $C \geq 1$.

For the other branch, let

$$J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

For a unit vector $x = (\xi, \eta)$, one has

$$\langle Jx, x \rangle = \bar{\xi}\eta, \quad \rho_J(x) = |\eta|^2.$$

Put $\mu = |\xi|$ and $\nu = |\eta|$, so that $\mu^2 + \nu^2 = 1$. The identities in (6) give

$$\frac{\sqrt{1 - r^2}}{r} = \frac{\kappa^2 - 1}{2\kappa}.$$

Therefore every point z in the disk indexed by x in (7) satisfies

$$|z| \leq \mu v + \frac{\kappa^2 - 1}{2\kappa} v^2 \leq \frac{\kappa}{2}.$$

The second inequality follows from

$$\frac{\kappa}{2} - \left(\mu v + \frac{\kappa^2 - 1}{2\kappa} v^2 \right) = \frac{(\kappa\mu - v)^2}{2\kappa} \geq 0. \quad (16)$$

Equality is attained by taking the unit vector $x = (\mu, v)$ with positive real coordinates

$$\mu = \frac{1}{\sqrt{\kappa^2 + 1}}, \quad v = \frac{\kappa}{\sqrt{\kappa^2 + 1}}.$$

For this vector, the positive boundary point is $z = \kappa/2$. Hence

$$\max_{z \in \Omega_r(J)} |z| = \frac{\kappa}{2}.$$

Since $\|J\| = 1$, taking the polynomial $f(z) = z$ yields

$$\frac{\|f(J)\|}{\max_{z \in \Omega_r(J)} |f(z)|} = \frac{2}{\kappa}.$$

Thus every admissible C satisfies $C \geq 2/\kappa$. Together with $C \geq 1$, this proves the claim. $\square$

5. Consequences and discussion

The transfer theorem accesses the base numerical-range estimate solely through (14), applied to the rank-one stretches (8) selected by the extraction mechanism. The disk-automorphism family in Lemma 3.1 supplies the orthogonal input-output pair used in (13); the identity function is one member of this family.

More generally, Theorem 4.1 turns every universal numerical-range spectral constant into an explicit constant for scaled q -numerical ranges.

For the sharp value $K = 2$, the transfer constant equals 1 precisely when $\kappa \geq 2$, or equivalently when $0 < |q| \leq 4/5$. At $|q| = 1$, one has $\kappa = 1$ and the argument reduces to the sharp numerical-range estimate. The definition of the scaled range uses $0 < |q| \leq 1$. When $n \geq 2$, unit vectors satisfying $y^*x = q$ exist for every such q .

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References

1. Crouzeix, M. Bounds for analytical functions of matrices. *Integral Equations and Operator Theory* **2004**, 48, 461–477. <https://doi.org/10.1007/s00020-002-1188-6>.
2. Jin, S. The numerical range is a 2-spectral set. Preprints.org, version 1, posted 27 July 2026, <https://doi.org/10.20944/preprints202607.1919.v1>.
3. Tsing, N.K. The constrained bilinear form and the C -numerical range. *Linear Algebra Appl.* **1984**, 56, 195–206. [https://doi.org/10.1016/0024-3795\(84\)90125-3](https://doi.org/10.1016/0024-3795(84)90125-3).
4. O’Loughlin, R.; Rani, J. q -numerical ranges and spectral sets. arXiv:2603.15536v1, submitted 16 March 2026, <https://doi.org/10.48550/arXiv.2603.15536>.